

How to customize the results of FMEDA on STM32 MCUs and MPUs

Introduction

This application note applies to the STM32 MCUs and MPUs and to X-CUBE-STL part number.

In the framework of STM32 MCUs and MPUs functional safety support package offer, several application notes reporting numeric results of safety analysis carried out on STM32 MCUs and MPUs are available. Such documents are usually indicated as “Results of FMEDA on microcontrollers of the STM32xx Series” (see document [2]). Because of the many STM32 MCUs and MPUs part numbers, the FMEDA results are delivered to end users in the form of static snapshots, computed for a given set of assumptions (see document [3]). The present application note provides clear instructions about how to customize such FMEDA numeric results in case end user application should require a different set of assumptions.





1 About this document

1.1 Purpose and scope

This document includes guidelines provided to customize the numeric outcomes reported in STM32 MCUs and MPUs FMEDA snapshots, according to specific end customer needs.

The process described in this document allows the end customers to virtually compute the FMEDA data for any STM32 MCU and MPU part number, without the need to ask ST sales to provide them. The associated approximation error is compatible with the scope and the purpose of FMEDA data provided in the framework of the STM32 functional safety support package.

The processes described in this document can also be used as a guidance to customize the FMEDA results related to STM8A MCU, in the framework of its functional safety support package.

This document applies to STM32 MCUs and MPUs Arm® Cortex® core-based devices.

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1.2 Terms and abbreviations

Table 1. Terms and abbreviations

Acronym	Definition
CoU	Conditions of use
CPU	Central processing unit
DC	Diagnostic coverage
FIT	Failure in time
FMEA	Failure mode effect analysis
FMEDA	Failure mode effect diagnostic analysis
ITRS	International technology roadmap for semiconductors
MCU	Microcontroller unit
MPU	Microprocessor unit
SFF	Safe failure fraction
SIL	Safety integrity level

1.3 Reference documents

- | | | |
|-----|------------------------------------|---|
| [1] | IEC/TR 62380:2004 | Reliability data handbook — Universal model for reliability prediction of Electronics components, PCBs and equipment |
| [2] | ANxxxx | Results of FMEDA on microcontrollers of the STM32xx Series (Example of AN5140: FMEDA snapshots for STM32F4 Series microcontrollers) |
| [3] | UMxxxx | STM32xx Series safety manual (Example of UM1840: STM32F4 Series safety manual) |
| [4] | IEC61508:1-7 [©] IEC:2010 | Functional safety of electrical/electronic/programmable electronic safety-related systems |
| [5] | AN5936 | X-CUBE-STL functional safety package: advanced topics |

2 How to compute FMEDA results for a different mission profile

The STMicroelectronics methodology for the application of the requirements specified by the functional safety standard IEC61508:2010 to the STM32 MCUs and MPUs adopts the international standard IEC62380:2004 as a reference for the computation of base failure rate for the component. This is clearly stated (in form of "assumption") in each FMEDA snapshot document as shown below for document [2]:

Caution:

The text in this section is not a replacement for IEC 62380 publication. Some equations are partially reported here with the mere scope of guidance for the end user during the FMEDA recomputation procedure. End user that intend to perform IEC 62380 computations need to access that publication.

Figure 1. FMEDA assumption on IEC62380 usage

Table 2. List of assumptions	
ID	Assumption
1	Base failure rates for permanent faults are computed on the basis of IEC62380 safety standard.
2	Internal debug-related structures are classified as "no effect", according to IEC61508-4, 3.6.4 ⁽¹⁾ .

Adopting IEC62380 means that the base failure rates for the different types of technology available in STM32 MCUs and MPUs (digital logic, nonvolatile memory, volatile memory, analog, package) are computed according to the mathematical model described in such safety standard (see document [1], section 7.3.1). In this particular case, the adopted mathematical model is the following:

$$\lambda = \left\{ \underbrace{\left\{ \lambda_1 \times N \times e^{-0.35 \times a} + \lambda_2 \right\} \times \left\{ \frac{\sum_{i=1}^y (\pi_i)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\}}_{\lambda_{die}} + \underbrace{\left\{ 2.75 \times 10^{-3} \times \pi_{\alpha} \times \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right) \times \lambda_3 \right\}}_{\lambda_{package}} + \left\{ \frac{\pi_I \times \lambda_{EOS}}{\lambda_{overstress}} \right\} \right\} \times 10^{-9} / h \quad (1)$$

The base failure rate is obtained as a sum of specific partial failure rates computations. According to the nature of STM32 MCUs and MPUs and considering the assumptions made in the related safety manual and FMEDA document, the device is not considered an interface. Accordingly, partial failure rate computation $\lambda_{overstress}$ is equal to zero (obtained by setting $\pi_1 = 0$ in above equation). Consequently, Eq. (1) provides the individual contributions for partial failure rates λ_{die} and $\lambda_{package}$ which can be managed separately.

The "mission profile" values affect the partial failure rates contributions in Eq. (1) computations, as defined in section 7.3.1 of IEC62380:2004 (see document [1], section 7.3.1). Furthermore, more explanations and "standard" combination of values representing reference mission profiles for different domains are reported in document [1], section 5.7). The affected portions of Eq. (1) by mission profile sections are either surrounded or strikethrough with blue lines in the figure below:

Figure 2. Equation sections of the model affected by mission profile settings

$$\lambda = \left\{ \underbrace{\left\{ \lambda_1 \times N \times e^{-0.35 \times a} + \lambda_2 \right\} \times \left\{ \frac{\sum_{i=1}^y (\pi_i)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\}}_{\lambda_{die}} + \underbrace{\left\{ 2.75 \times 10^{-3} \times \pi_{\alpha} \times \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right) \times \lambda_3 \right\}}_{\lambda_{package}} + \left\{ \frac{\pi_I \times \lambda_{EOS}}{\lambda_{overstress}} \right\} \right\} \times 10^{-9} / h$$

The FMEDA snapshots for STM32 MCUs and MPUs are computed for a specific reference mission profile (usually “Telecom - Ground fixed switching” is used). The adopted mission profile is always clearly indicated in the FMEDA document (see blue rectangle below):

Figure 3. Mission profile adopted

5.1 STM32F469IIT	
Part number	STM32F469II
Mission profile	Telecom - Ground fixed switching
SER data source (transient faults)	ITRS2011, sea level
Fault injection results included	No

In case a different mission profile is required by the end user, failure rates and relative safety metrics need to be recomputed, so the highlighted parts in Figure 2 must be reconsidered. To do this, use the definition of the following subset of Eq. (1), where Mpf acronym means “Mission profile Factor”:

$$\text{Mpf}(\text{die}) = \left\{ \frac{\sum_{i=1}^v (\pi_t)_i \times \tau_i}{\tau_{on} + \tau_{off}} \right\} \quad \text{Mpf}(\text{package}) = \left(\sum_{i=1}^z (\pi_n)_i \times (\Delta T_i)^{0.68} \right) \quad (2)$$

Eq. (1) shows that each individual failure rate for die or package is multiplied by the related Mpf value. In other words, the Mpf is a “common factor” for all failure rates of the component related to concrete mission profile definition. When mission profile is changed, the user needs to calculate new Mpf values. Then, to get the new set of values describing failure rates included in FMEDA static snapshots related to a modified mission profile, it is enough to “scale” the given reference failure rate by the ratio between two Mpf (the reference one, and the new one) as in the following equation:

$$\lambda(\text{new mission profile}) = \lambda(\text{reference mission profile}) \times \frac{\text{Mpf}(\text{new mission profile})}{\text{Mpf}(\text{reference mission profile})} \quad (3)$$

The reference Mpf values used for FMEDA computation in STMicroelectronics documents are indicated at the end of each section related to a specific device, as shown below for document [2]:

Figure 4. Reference mission profile indication in FMEDA documents

Table 7. Mission profile factors - STM32F469IIT	
Die	1.14
Package	364.29

The above values are Mpf(die) and Mpf(package) used during FMEDA computation in order to generate the outcomes for the specific device.



Because FMEDA static snapshots share the same structure (that is, the same sequence of four numeric tables), here below is reported the list of modification to be applied to get the new complete set of safety metrics with the new mission profile setting.

First table (see [2], example document, Table 3): Safety results (MCU level) - Absolute metrics

Permanent failure rates for “Die only” and “Package” need to be computed according to Eq. (3).

Permanent failure rates for “SOC total” need to be recomputed (they are the sum of “Die only” and “Package” contribution)

Transient failure rates are unchanged (not affected by mission profile change)

Second table (see [2] example document, Table 4): Safety results (MCU level) - Relative metrics

Safety metrics for permanent faults related to “Die only” and “Package” are unchanged (mission profile modification affects in the same way permanent failure rates, so the computations for relative safety metrics give the same result).

Safety metrics for permanent faults related to “SOC total” need to be computed. As the DC is defined as (according to IEC61508-2:2010, Annex C):

$$DC = \frac{\lambda_{DD}}{\lambda_{DD} + \lambda_{DU}} \quad (4)$$

Therefore, because overall λ for SOC is composed by the contribution of die and package, we have:

$$DC(SOC) = \frac{\lambda_{DD}(\text{die}) + \lambda_{DD}(\text{package})}{\lambda_{DD}(\text{die}) + \lambda_{DU}(\text{die}) + \lambda_{DD}(\text{package}) + \lambda_{DU}(\text{package})} \quad (5)$$

DC for permanent faults can be therefore computed with the above formula. Similar considerations apply to SFF, which is defined this way:

$$SFF = \frac{\lambda_{DD} + \lambda_{SAFE}}{\lambda_{DD} + \lambda_{DU} + \lambda_{SAFE}} \quad (6)$$

Safety metrics for transient faults are unchanged (not affected by mission profile change)

Third table (see [2] example document, Table 5): Safety results (by function) - Absolute metrics

Permanent failure rates need to be computed according to Eq. (3).

Transient failure rates are unchanged (not affected by mission profile change)

Fourth table (see [2] example document, Table 6): Safety results (by function) - Relative metrics

Safety metrics for permanent faults are unchanged (mission profile modification affects in the same way permanent failure rates, so the computations for relative safety metrics give the same result).

Safety metrics for transient faults are unchanged (not affected by mission profile change)

Note: According to this procedure, the computation of new Mpf (mission profile factor) values to be used to obtain updated FMEDA outcomes is under end user responsibility. The Mpf(package) requires the computation of thermal amplitude variation of the mission profile, which can be obtained by exploiting thermal-related data (for example, Θ_{JA} - junction to ambient thermal resistance) included in the specific device datasheet.

Note: In principle there are no accuracy errors associated with the computations described above in order to adjust FMEDA outcomes to a new mission profile – the procedure is entirely based on correct mathematic steps with no approximations. End users might use this application note and the related rational to show during an assessment phase the correctness of the new obtained data.



3 How to compute FMEDA results for a different package type

FMEDA snapshot data are computed for a specific STM32 MCU or MPU part number, and so for a specific package. Among a given STM32 MCU or MPU series, several different packages are available. This section provides a guideline to recompute FMEDA metrics with a package different from the one used for the snapshot. First, identify the package used for the available snapshot. Pay attention, if several FMEDA snapshots are available for the given STM32 series, select the new ones with the peripheral combination nearest to the part number required. Then, identify the new package for which the FMEDA metrics are needed.

According to IEC62380 model used for base failure rate computations (refer to for details) the base failure rate for different packages differs just by a scaling factor (in the specific, λ_3 factor directly depends on the physical dimension of the package). Following reference table provides a way to compute the relationship between the available safety metrics, and the new ones. It is assumed that the DC and safeness associated to the package are still the same used in the FMEDA snapshot; if this is not the case, further computations can embed the new assumptions.

Table 2. Scaling factor F for packages

Package	F
LQFP32	0.549
LQFP48	0.549
LQFP64	1.000
LQFP80	1.760
LQFP100	1.760
LQFP128	1.760
LQFP144	3.204
LQFP176	4.353
LQFP208	5.639
UFQFPN28	0.215
UFQFPN48	0.549
VFQFPN32/UFQFPN32	0.312
TSSOP20	0.220
UFBGA132	0.835
UFBGA169	0.835
UFBGA176+25	1.000
TFBGA100	1.045
TFBGA216	2.363
TFBGA240+25	2.677
WLCSP25	0.070
WLCSP81	0.368
WLCSP144	0.507
WLCSP168	0.507
WLCSP180	0.645

The new values for λ (absolute safety metrics in FIT) must be computed with this formula by using “F” values extracted from above table

$$\lambda(\text{newpackage}) = \lambda(\text{oldpackage}) \cdot \frac{F(\text{newpackage})}{F(\text{oldpackage})} \quad (7)$$

Then, the FMEDA overall data must be recomputed according to the following guidelines:

**First table (see [2], example document, Table 3): Safety results (MCU level) - Absolute metrics**

All permanent failure rates for “Package” need to be computed according to Eq. (7).

Permanent failure rates for “Die only” are unchanged (not affected by package change)

Permanent failure rates for “SOC total” need to be recomputed (they are the sum of “Die only” and “Package” contribution)

Transient failure rates are unchanged (not affected by package change)

Second table (see [2] example document, Table 4): Safety results (MCU level) - Relative metrics

Apply the same computation pattern used for this table in How to compute FMEDA results for a different mission profile

Third table (see [2] example document, Table 5): Safety results (by function) - Absolute metrics**Fourth table (see [2] example document, Table 6): Safety results (by function) - Relative metrics**

Unchanged (not affected by package change)

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4 How to compute FMEDA results for a different memory size

FMEDA snapshot data are computed for a specific STM32 MCU or MPU part number, and for a specific combination of SRAM and flash memory sizes. STM32 devices features, in the same series, different memories sizes, leading to the need to recompute FMEDA snapshot data accordingly.

A simplified process to recompute FMEDA data is available, and it is based on two main assumptions:

1. Memory absolute failure rates (λ s) have a linear dependence on memory size. Refer to [Note 1].
2. Memory relative failure rates (DC, SPFM) do not depend on memory size. Refer to [Note 2].

The formula to be used for the computation of updated values of λ is the following:

$$\lambda(\text{newmemorysize}) = \lambda(\text{oldmemorysize}) \cdot \frac{\text{Size}(\text{newmemorysize})}{\text{Size}(\text{oldmemorysize})} \quad (8)$$

Computation must be done for all λ s of the changed memory (both permanent and transient). Accordingly, the overall FMEDA data must be recomputed according to these guidelines:

Third table (see [2] example document, Table 5): Safety results (by function) - Absolute metrics

Permanent and transient failure rates for the memory (or memories) that changed size must be recomputed according to Eq. (8).

Fourth table (see [2], example document, Table 6): Safety results (by function) - Relative metrics

Unchanged (not affected by memory size change according to the assumptions reported above).

First table (see [2], example document, Table 3): Safety results (MCU level) - Absolute metrics

Permanent and transient failure rates for “Die only” must be recomputed (they are the sum of all individual failure rates of modules and so affected by the change of memory/memories failure rates change).

Permanent failure rates for “Package” are unchanged (not affected by memory size change).

Permanent and transient failure rates for “SOC total” need to be recomputed (they are the sum of “Die only” and “Package” contribution).

Second table (see [2], example document, Table 4): Safety results (MCU level) - Relative metrics

Apply the same computation pattern used for this table in How to compute FMEDA results for a different mission profile.

Notes	
[Note 1]	This is usually true for transient failure rates. For permanent failure rates, applying the simplified procedure is, in principle, not entirely correct because the FMEDA computation process includes some nonlinearities. These nonlinearities are related to the different contributions in FMEDA data for memories (memory cells and address decoder), and to the nonlinear model for base failure rate calculation in reference [1]. However, considering that: • the IEC 62380 nonlinear model is generally considered very conservative • the recommended safety mechanism combinations are assumed to be implemented correctly • the resulting λDU for permanent failures of flash memory and RAM are orders of magnitude lower than the absolute targets for SIL2 safety functions, then the simplified procedure can be still considered applicable. See reference [5] for further details on the overall strategy related to base failure rates within the STM32 safety concept.
[Note 2]	This is true for homogeneous groups of memories. Because FMEDA data often combines all memories (for example, grouping all SRAMs), in case a subset of the memories features different combination of safety mechanisms (for example, one SRAM bank features ECC protection, and the others not) the simplified FMEDA recomputation proposed in this section cannot be done. In such a case, one conservative way is to decrease the associated DC down to 90% (Medium), and to recompute SRAM λDUs following the procedure of Section 5.



5 How to compute FMEDA results with different combinations of safety mechanisms

According to the document [3] and FMEDA snapshot documents, FMEDA data are computed assuming that all safety mechanisms rated as “Highly recommended” (++) are correctly implemented and active.

In the case that the safety mechanisms have not been correctly implemented or are not active, FMEDA data for impacted MCU modules must be recomputed because of changes to the DC value associated with impacted modules. To recompute the FMEDA data, use the following procedure:

- From the FMEDA tables, retrieve the original module dangerous failure rate $\lambda D = \lambda DU + \lambda DD$.
- Apply the new DC value to get the updated λDU and λDD ($\lambda DD = DC \lambda D$; $\lambda DU = (1 - DC) \lambda D$).
- Recompute the entire MCU failure as one or more modules has/have been changed.

Note: *The procedure must be done separately for both permanent and transient failure rates.*

The value to be used for the new DC must be decided by the end user, and is the end user's responsibility. To maintain coherence with the current STM32 safety concept, it is always recommended to link any decision on the achieved DC to the indication provided by [4], Annex A, tables A.2 to A.14.

Note: *The strategy for FMEDA document modification (which tables need to be updated) is the same as described in Section 3.*

Important: This method is always admissible when additional layers of protection are added to the safety mechanism combination assumed in FMEDA data, therefore resulting in a DC value greater than what actually used in the published FMEDA. If, instead, one or multiple safety mechanisms rated ++ are skipped, potential serious consequences (extending beyond the DC decrease for impacted modules) to the STM32 MCU/MPU safety concept may arise. Refer to document [3] to understand the real role of each safety mechanism in the safety concept.



6 How to compute FMEDA results with a different base failure rate

As clarified in [Figure 1. FMEDA assumption on IEC62380 usage](#), and further developed in reference [5], FMEDA computations are based on specific assumed base failure rates for permanent and transient failures. There could be reasons for the end user to recompute absolute safety metrics by using a different base failure rate: for instance, to use an alternative reliability model (like the SN29000) instead of IEC62380, or to scale transient failure rates with the altitude.

Permanent failures

The end user must execute the procedure below, according to selected fault model:

1. To calculate the new absolute failure rate for the whole MCU according to the new standard/reliability model
2. To calculate the failure distribution of the selected STM32 modules by using the absolute permanent failure rate $\lambda_{SR} = \lambda_{DD} + \lambda_{DU} + \lambda_{SAFE}$ (from [2] example document, Table 5: Safety results (by function) - Absolute metrics)
3. To distribute the whole MCU failure rate obtained in step 1 by using the distribution obtained in step 2

Note:

In most cases, reliability models/books do not provide separate failure rates for die and package. The partitioning available in FMEDA documents, based on the IEC 62380 models, can be assumed as accurate and reused as it is.

Transient failures

The end user must execute the procedure below:

1. To calculate the new base failure rate for bit according to the new transient failures model/data (*)

Note:

The STMicroelectronics quality department can provide SER figures coming from experimental tests, also including a simplified table to scale up SER data compared to altitude on the basis of Annex A of JESD89A. Contact your local STMicroelectronics sales representative to access to these quality reports.

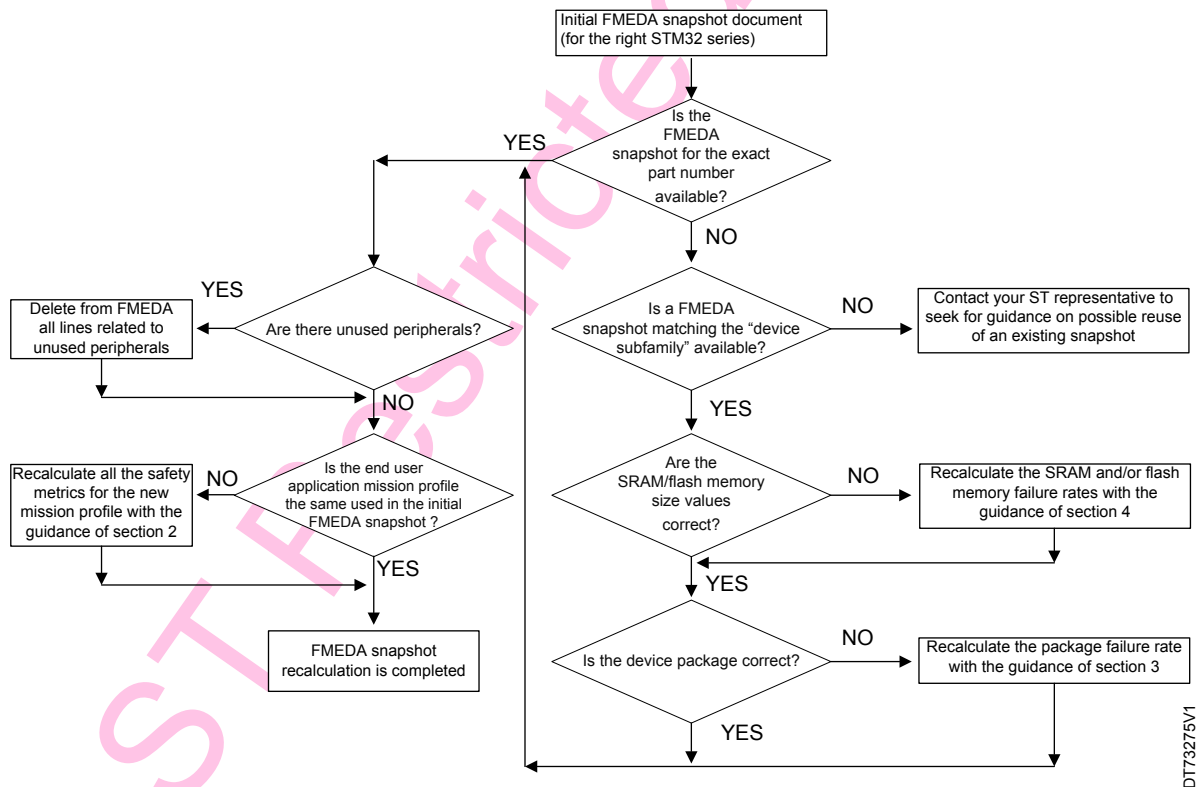
2. To identify the value used in FMEDA document for transient base failure rate, usually reported in [Figure 1. FMEDA assumption on IEC62380 usage](#) (if not, refer to [5] for further details on the used value.)
3. To scale all transient failure rates accordingly, by using the ratio between the two above base failure rates



7 How to derive a FMEDA snapshot for a non-supported part number

FMEDA data are provided in form of static snapshots for a limited quantity of supported partnumbers. In case the snapshot for the part number needed by the end user is not included in the available FMEDA documents for its STM32 series, the procedure of this section allows – under specific conditions – to obtain the needed snapshot by combining several methods already explained in another section of the document.

Figure 5. How to derive a FMEDA snapshot for a non-supported partnumber



DT73275V1



Revision history

Table 3. Document revision history

Date	Version	Changes
29-Apr-2019	1	Initial release
10-Sep-2020	2	Updated: Purpose and scope Added: - How to compute FMEDA results for a different package type - How to compute FMEDA results for a different memory size
31-Aug-2023	3	Updated: Section 3 How to compute FMEDA results for a different package type Section 4 How to compute FMEDA results for a different memory size Added: Reference [5] Section 5 How to compute FMEDA results with different combinations of safety mechanisms Section 6 How to compute FMEDA results with a different base failure rate Section 7 How to derive a FMEDA snapshot for a non-supported part number



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